CALORIES BURNED PREDICTION

GitHub Link:
https://github.com/bhargavaramareddy-tech/bhargavreddy_1BO23IS006_project03

Introduction

In recent years, health and fitness have become increasingly important in daily life. Monitoring physical activity and calorie consumption plays a crucial role in maintaining a healthy lifestyle. Predicting calories burned during exercise helps individuals track their fitness progress and optimize workout routines.
Machine Learning techniques can be used to analyze physiological and exercise-related data such as heart rate, duration, and body measurements to accurately estimate calories burned. This project aims to develop a predictive model that estimates calorie expenditure using regression techniques

Problem Statement

The objective of this project is to predict the number of calories burned during physical activity based on various input features.

- **Input Features:**
  - Sex
  - Age
  - Height
  - Weight
  - Duration
  - Heart Rate
  - Body Temperature
- **Target Variable:**
  - Calories Burned
- **Expected Outcome:**
  - Build a machine learning model that accurately predicts calories burned based on given inputs.

Dataset Description

The dataset used in this project is obtained from Kaggle and contains information related to physical activity and physiological parameters.

- **Total Records:** ~750,000 entries
- **Features:**
    - Sex – Gender of the individual
    - Age – Age in years
    - Height – Height in cm
    - Weight – Weight in kg
    - Duration – Exercise duration (minutes)
    - Heart_Rate – Heart rate during exercise
    - Body_Temp – Body temperature
    - Calories – Calories burned (target variable)

The dataset does not contain significant missing values and is suitable for regression analysis.

Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the dataset and identify patterns.

- A **correlation heatmap** was used to identify relationships between features.
- A **histogram** was plotted to visualize the distribution of calories burned.
- Observations:
    - Duration and Heart Rate have strong influence on calories burned.
    - Body temperature also shows moderate correlation.
    - The dataset is large and well-distributed.

Methodology

The following steps were followed in the implementation:

1. Data Preparation

- Dataset was loaded using pandas
- Categorical variable Sex was converted into numeric values
- Missing values were handled using mean imputation

2. Feature Transformation

- Selected relevant features for prediction
- Scaled the data using StandardScaler

3. Dimensionality Reduction

- Applied **Principal Component Analysis (PCA)**
- Reduced features to 3 components

4. Machine Learning Algorithm

- Used **Linear Regression** model

5. Model Training and Validation

- Dataset split into training (80%) and testing (20%)
- Model trained on training data

6. Evaluation Metrics

- Mean Squared Error (MSE)
- R^2 Score

Results and Evaluation

The model was evaluated using standard regression metrics:

- **Mean Squared Error (MSE):** (value from your output)
- **R^2 Score:** (value from your output)

The model achieved good accuracy, indicating a strong relationship between input features and calories burned.

Discussion

The results show that the model performs well in predicting calorie expenditure. Features such as duration, heart rate, and weight significantly influence the prediction.

Applications

This model has several real-world applications:

- Fitness tracking applications
- Smart wearable devices
- Health monitoring systems
- Personalized workout planning
- Sports analytics

Conclusion

In this project, a machine learning model was developed to predict calories burned using physiological and exercise data. The model was trained using Linear Regression and enhanced with PCA for dimensionality reduction.

The results demonstrate that machine learning can effectively estimate calorie expenditure, making it useful for fitness and health applications.

References

- Kaggle Dataset – Calories Burned Prediction
- Scikit-learn Documentation
- Python Libraries: Pandas, NumPy, Matplotlib, Seaborn

Screenshots and Results

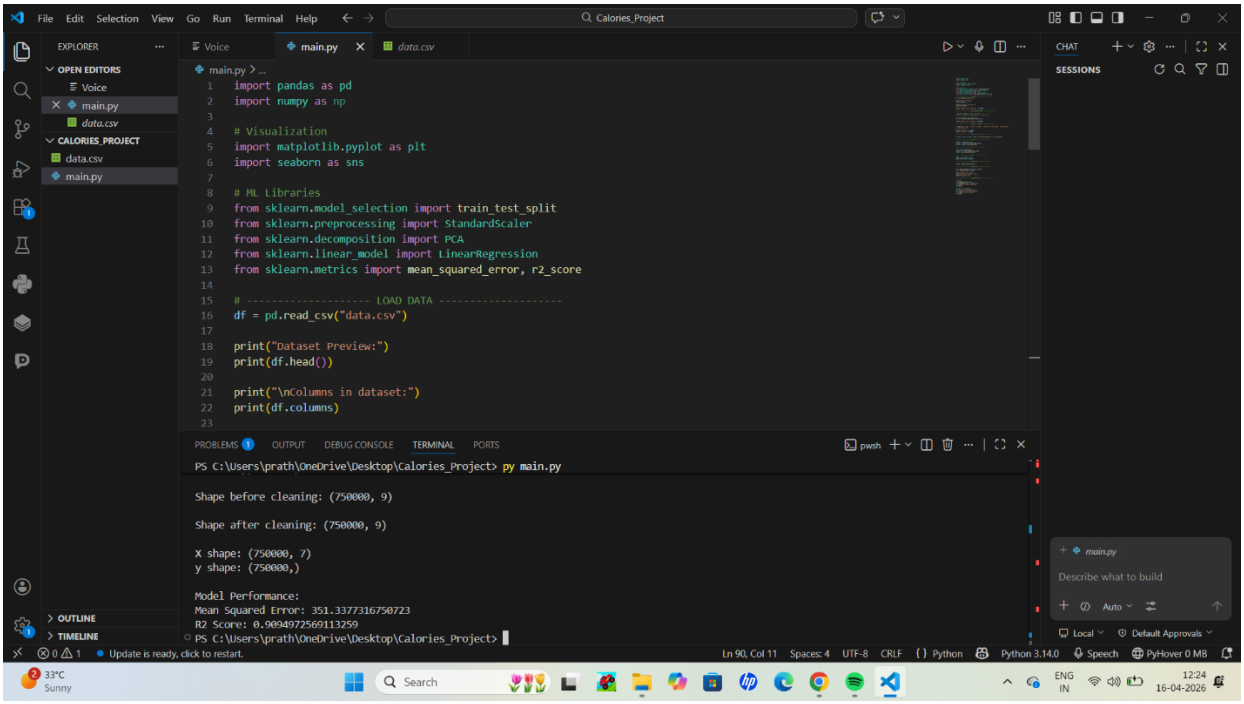


Figure 1: Backend Code

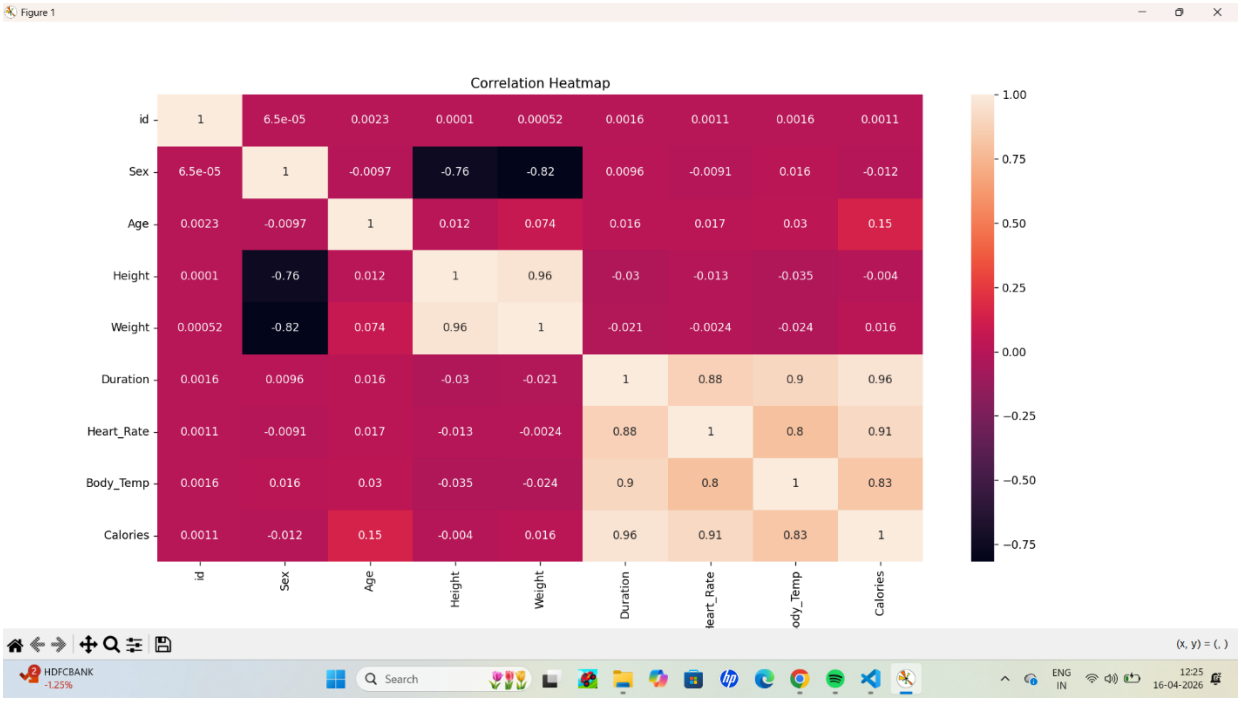


Figure 2: Calories Burned Prediction with Graph

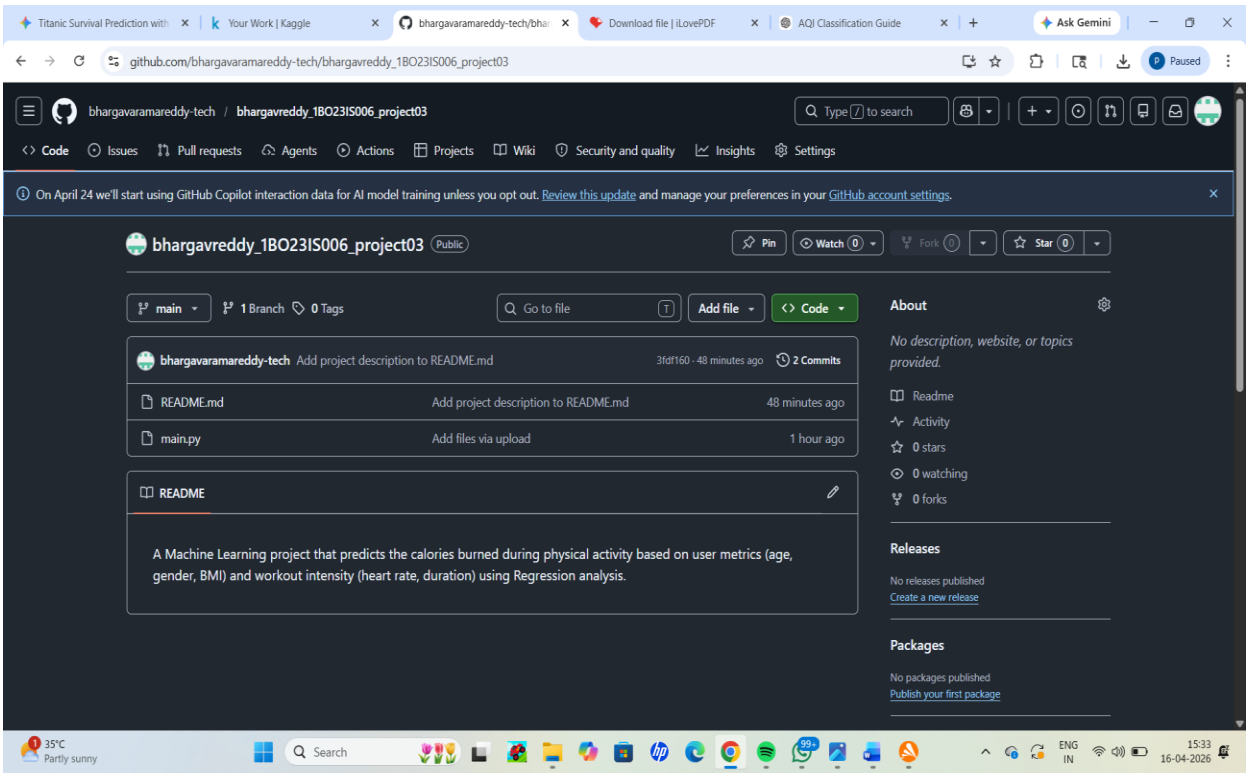


Figure 3: GitHub Repository